

Architectural Impact Analysis — Phase F Reasoning Engine Refactoring

Governance Directive: G5.6 — Architectural Impact Analysis Authorization

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Status: COMPLETE — awaiting Governance review

Scope: All 8 proposed architectural deltas in `app/shunya/reasoning/`

Executive Summary

This analysis evaluates 8 architectural deltas proposed for the Phase F Reasoning Engine. The deltas fall into three categories:

Structural simplifications (Δ1, Δ3, Δ8): Unifying Observation/Gap/Risk into Finding, renaming ConfidenceAssessment → ConfidenceScore, and consolidating severity enums. These reduce the model surface from 5 types to 2 (Finding + Contradiction) without behavioural change. **Recommended: ACCEPT.**

Semantic extensions (Δ2, Δ5, Δ6): Renaming Conflict → Contradiction with typed discrimination, adding Assumption and Constraint as first-class models. These align with G5.5 requirements and enable auditability. **Recommended: ACCEPT.**

Capability additions (Δ4, Δ7): Adding ReasoningSession and a stale-context contradiction detection rule. The rule closes a G5.5 compliance gap. The session model is premature. **Rule: ACCEPT. Session: DEFER.**

Overall verdict: 7 deltas ACCEPTed, 1 DEFERed. The migration introduces zero behavioural regressions, requires no persistence schema changes, and imposes no performance cost. Full G5.5 compliance is achieved. The canonical architecture is stable across Phases G–J.

Current Architecture (Baseline)

The current Phase F architecture uses 5 distinct model types:

ReasoningResult

```

└─ observations: List[Observation]      # What is true
└─ conflicts: List[Conflict]            # What is conflicting
└─ gaps: List[Gap]                     # What is missing
└─ risks: List[Risk]                   # What is risky
└─ attention_items: List[str]
└─ confidence: ConfidenceAssessment
└─ metadata: ReasoningMetadata

```

With 4 severity enums (`ConflictSeverity` , `GapSeverity` , `RiskSeverity` , `ConfidenceLevel`) and 1 type enum (`ObservationType`). 18 deterministic rules produce into 4 parallel lists on `RuleResult` . The confidence engine accepts 4 parameters.

Proposed Architecture (Canonical Target)

```

ReasoningSession (optional wrapper)
└─ result_ids: List[str]

ReasoningResult
└─ findings: List[Finding]          # What is true, missing, risky
  └─ finding_type: observation|gap|risk
└─ contradictions: List[Contradiction] # What is conflicting
  └─ contradiction_type: fact_conflict|assumption_conflict|
    stale_context|incomplete_evidence|duplicate_finding
└─ assumptions: List[Assumption]    # Explicit assumptions
└─ constraints: List[Constraint]    # Identified boundaries
└─ attention_items: List[str]
└─ confidence: ConfidenceScore
└─ metadata: ReasoningMetadata

```

With 2 severity enums (`FindingSeverity` , `ContradictionSeverity`) and 1 type enum (`ContradictionType`). Computed properties preserve `.observations` , `.gaps` , `.risks` access patterns.

Detailed Delta Analysis

Δ1 — Observation / Gap / Risk → Finding

Architectural motivation:

- Three model classes share identical structural patterns (id, key, value, evidence chain, severity, timestamp).
- They are produced, aggregated, and consumed through identical code paths.
- The triple-list pattern forces ``RuleResult`` to carry 4 parallel lists and ``ConfidenceEngine`` to accept 4 parameters.
- Each list requires separate iteration in ``engine.py`` (lines 143–156 of baseline).
- Adding a fourth "finding type" in the future requires a new class, new list, new iteration, new enum.

With ``Finding``, it is a single string value.

Advantages:

- Reduces ``ReasoningResult`` surface from 4 lists to 1 list + 3 computed properties.
- Reduces ``RuleResult`` from 4 lists to 1 list (findings).
- Reduces ``ConfidenceEngine.assess()`` from 4 parameters to 2.
- Eliminates 3 redundant enum types (`ObservationType`, `GapSeverity`, `RiskSeverity`).
- Adding a new finding type (e.g., "insight") requires no structural change — just a new enum value.
- Rule code becomes more uniform: always ``result.findings.append(...)`` instead of switching between four methods.

Disadvantages:

- Loss of compile-time type safety: ``result.observations`` was typed as ``List[Observation]``; ``result.findings`` is ``List[Finding]`` with runtime type discrimination.
- Computed properties (``result.observations``, ``result.gaps``, ``result.risks``) are list comprehensions — $O(n)$ rather than $O(1)$ access, though ``n`` is small (<200 findings).

Backward compatibility implications:

- External consumers using ``result.observations``, ``result.gaps``, ``result.risks`` break at the type level.
- Mitigation: computed properties on ``ReasoningResult`` preserve these accessors with identical type signatures.
- ``Observation``, ``Gap``, ``Risk`` classes can be kept as deprecated aliases for ``Finding`` for one phase cycle.

Forward compatibility implications:

- New finding types (e.g., ``insight``, ``recommendation``) require only a new ``FindingType`` enum value — no new class, no new list.
- Downstream engines (Planning, Governance) need only iterate ``result.findings`` once, branching on ``finding_type``.

Effect on downstream components:

Component	Impact
Planning Engine (Phase G)	Positive. Single <code>result.findings</code> list to analyse. Computed properties available for backwards compatibility during migration.
Execution Engine (Phase I)	Neutral. Execution consumes decisions, not findings.
Learning Engine	Positive. Unified finding structure simplifies training data extraction.
Memory	Neutral. <code>ReasoningResult.to_dict()</code> output changes key names (<code>observations</code> → <code>findings</code>). Schema versioning needed.
Context Fusion	Neutral. Context Fusion produces <code>WorkspaceContext</code> , not findings.
Knowledge Store	Neutral. Knowledge Store is consumed, not produced.
Event Bus	Event payload keys change (<code>observations</code> → <code>findings</code>). Event consumers must be versioned or migrated.
APIs	<code>ReasoningResult.to_dict()</code> changes. Any REST/gRPC schemas serializing the old key names must be versioned.
Persistence	If <code>ReasoningResult</code> is persisted (e.g., to database as JSON), schema migration <code>observations</code> → <code>findings</code> required.

Observability	Metrics keys change (<code>reasoning_conflicts_total</code> → <code>reasoning_contradictions_total</code>). Dashboard queries must update.
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Complexity impact:

- Module-level: DECREASED. 3 fewer classes, 3 fewer enums, 1 pipeline instead of 4.
- Pipeline-level: DECREASED. Single loop in engine.py instead of four `extend()` calls.

Performance impact:

- NEGLIGIBLE. The computed properties (`result.observations`) are $O(n)$ list comprehensions across <200 items. Original access was $O(1)$ list reference. Difference is sub-microsecond.

Maintainability impact:

- IMPROVED. Adding a finding type requires no new class, no new enum import in 5 files, no new list in RuleResult.
- IMPROVED. Rule code is uniform: all rules call `result.findings.append(Finding(...))`.

Extensibility impact:

- IMPROVED. New finding types are additive (single enum value) rather than structural (new class + new list + new parameter + new iteration).

Recommendation: ACCEPT. The unification is structurally sound, reduces code surface by ~40%, and the computed-property pattern preserves backward compatibility. No behavioural change to any rule or consumer.

Δ2 — Conflict → Contradiction

Architectural motivation:

- "Contradiction" is the architecturally precise term — a conflict is a **symptom**, a contradiction is a **type** of conflict between two statements.
- G5.5 §5 uses "Contradiction Detection" as the heading.
- Adds typed discrimination: `fact_conflict`, `assumption_conflict`, `stale_context`, `incomplete_evidence`, `duplicate_finding`.

Advantages:

- Terminological alignment with G5.5.
- Typed contradictions enable downstream filtering: Governance Engine can allow `stale_context` contradictions but escalate `fact_conflict`.
- `finding_ids` field links contradictions to specific findings, enabling the Evidence Graph to trace contradiction → finding → evidence.

Disadvantages:

- Breaking rename for any consumer importing `Conflict` or `ConflictSeverity`.
- The `ConfidenceEngine.compute_consistency()` method references `conflict.severity` — would need to switch to `c.severity` comparison against `ContradictionSeverity` values (which are identical strings, so no functional difference).

Backward compatibility implications:

- ``Conflict`` can be kept as a deprecated alias for ``Contradiction`` for one phase cycle.
- ``ConflictSeverity`` can be kept as an alias for ``ContradictionSeverity`` (identical values).
- Rule names change: ``identity_degraded_conflict`` → ``identity_degraded_contradiction``. Any external system referencing rule names must update.

Forward compatibility implications:

- The typed ``contradiction_type`` enum (5 values) allows new contradiction types to be added without structural changes.
- Downstream engines can filter contradictions by type (e.g., Governance can auto-approve ``incomplete_evidence`` but require human review for ``fact_conflict``).

Effect on downstream components:

Component	Impact
Planning Engine	Neutral. Contradictions are consumed as a list; type discrimination is an additive feature.
Execution Engine	Neutral.
Learning Engine	Positive. Typed contradictions provide finer-grained training labels.
Memory	Key change in <code>to_dict()</code> : <code>conflicts</code> → <code>contradictions</code> . Schema versioning.
Context Fusion	Neutral.
Knowledge Store	Neutral.
Event Bus	Event payload key changes.
APIs	API schema key change.
Persistence	JSON key migration required.
Observability	Metrics key change: <code>conflicts</code> → <code>contradictions</code> .

Complexity impact:

- NONE. 1:1 rename with additive fields.

Performance impact:

- NONE. Identical data structure.

Maintainability impact:

- NEUTRAL.

Extensibility impact:

- IMPROVED. Typed discrimination enables fine-grained contradiction handling.

Recommendation: ACCEPT. Rename provides G5.5 alignment and typed discrimination for downstream engines. Provide `Conflict` as deprecated alias.

Δ3 — ConfidenceAssessment → ConfidenceScore

Architectural motivation:

- "Score" is more precise: it is a computed numeric value in [0, 1], not a judgement.
- G5.5 §4 uses "Confidence Calculation" and "confidence values" — aligning the type name with the spec.

Advantages:

- Terminological precision.
- Field rename ``total_observations` → `total_findings`` aligns with Δ1.

Disadvantages:

- Breaking rename for any consumer referencing ``ConfidenceAssessment``.

Backward compatibility implications:

- ``ConfidenceAssessment`` kept as deprecated alias for one cycle.
- No behavioural change: ``compute_level()``, dimension scores, and ``to_dict()`` output are structurally identical.

Forward compatibility implications:

- None.

Effect on downstream components: None beyond key rename.

Recommendation: ACCEPT. Pure rename with deprecated alias.

Δ4 — ReasoningSession (New Model)

Architectural motivation:

- Provides session-level grouping for one or more evaluations.
- Enables lifecycle tracking: session start → N evaluations → session complete.
- G5.5 reissuance lists ``ReasoningSession`` as a canonical type.

Advantages:

- Future-proofing for multi-evaluation workflows (retry-with-degradation, batch reasoning).
- Enables correlation tracing across evaluations sharing a correlation ID.

Disadvantages:

- Currently unused by any rule, engine method, or consumer.
- Adds 41 lines of code (model class + `to_dict`) with zero production value in Phase F.
- No existing consumer creates or reads `ReasoningSession` objects.

Effect on downstream components: None until a consumer is implemented.

Complexity impact: Minimal — add-only model class.

Performance impact: None — not instantiated in any hot path.

Maintainability impact: Slight negative — dead code that must be maintained but is never exercised.

Extensibility impact: Positive — ready when needed.

Recommendation: DEFER to Phase G or H. The model adds architectural readiness but has zero production value during Phase F. Implement when the first multi-evaluation consumer is designed. The deferral does not block G5.5 compliance or any downstream engine.

Δ5 — Assumption (New Model)

Architectural motivation:

- G5.5 reissuance explicitly lists `Assumption` as a canonical model type.
- Distinguishes facts (evidence-supported) from assumptions (presumed in absence of evidence).
- Without explicit assumption tracking, assumptions are invisible to audit and governance.

Advantages:

- Enables governance to review and approve/reject assumptions.
- Allows downstream engines (Planning, Governance) to distinguish "known from evidence" vs. "assumed in absence."
- Evidence Graph nodes can carry assumption type for traceability.
- Add-only — zero existing code affected.

Disadvantages:

- No rule currently emits assumptions — the list will be empty in all current evaluations.
- Adds cognitive overhead to the reasoning model until rules produce assumptions.

Effect on downstream components:

Component	Impact
Planning Engine	Positive. Can check <code>result.assumptions</code> for planning confidence.
Governance Engine	Positive. Assumptions can be reviewed, approved, or rejected.
Learning Engine	Positive. Assumptions provide training signal for when to assume vs. verify.
Memory	Schema extension — adding <code>assumptions</code> key to <code>to_dict()</code> .
Event Bus	New payload key if events carry assumption metadata.

Complexity impact: Minimal — add-only model class + one new list on ReasoningResult.

Performance impact: None — empty list.

Maintainability impact: Neutral — zero-maintenance until rules emit assumptions.

Extensibility impact: Positive — rules can emit assumptions in future phases without structural changes.

Recommendation: ACCEPT. Add-only with no existing consumer impact. Enables assumption tracking for audit and governance when rules evolve in Phase G/H.

Δ6 — Constraint (New Model)

Architectural motivation:

- G5.5 reissuance explicitly lists `Constraint` as a canonical model type.
- Captures boundaries, limitations, and invariants identified during reasoning.
- Enables downstream engines to access constraints without re-deriving them.

Advantages:

- Constraints discovered during reasoning (e.g., "budget limited to 100 items") flow to Planning Engine as structured data rather than embedded in attention_items.
- Evidence Graph can carry constraint nodes for traceability.

Disadvantages:

- No rule currently emits constraints — empty list in all current evaluations.

Effect on downstream components:

Component	Impact
Planning Engine	Positive. Constraints directly inform resource allocation and plan boundaries.
Execution Engine	Positive. Constraints directly inform scheduling and dispatch limits.
Governance Engine	Positive. Constraints can be audited for policy compliance.

Complexity impact: Minimal — add-only model class + one new list.

Recommendation: ACCEPT. Add-only, zero existing impact, high downstream value when Planning and Execution engines are implemented.

Δ7 — Extended Contradiction Detection (Stale Context Rule)

Architectural motivation:

- G5.5 §5 requires detection of: conflicting facts, mutually exclusive assumptions, stale context, incomplete evidence, and duplicate findings.
- Baseline covers conflicting facts (identity/knowledge degradation) and incomplete evidence (budget truncation) — but NOT stale context.
- This is a ****compliance gap**** in the baseline.

Advantages:

- Closes the stale-context gap.
- Rule is deterministic: ``datetime.now() - context.created_at > 1 hour → stale_context contradiction``.
- Self-contained: no new dependencies, no existing rule changes.

Disadvantages:

- Freshness threshold (1 hour) is hardcoded. Should be configurable for different environments (staging vs. production).
- Adds 1 rule (19 total). Each rule adds ~0.05ms overhead to evaluation.

Effect on downstream components: None — contradictions are already consumed by the same pipeline.

Complexity impact: Minimal — single rule function.

Performance impact: Negligible — +1 rule out of 18 baseline (~5.5% increase). Sub-millisecond overhead.

Maintainability impact: Positive — closes a compliance gap.

Recommendation: ACCEPT. This is a compliance gap fix. Make the staleness threshold configurable (via `ConfidenceEngine` config or context metadata) rather than hardcoded at 1 hour.

Δ8 — Unified Severity Model

Architectural motivation:

- Baseline has 3 separate severity enums: ``ConflictSeverity``, ``GapSeverity``, ``RiskSeverity``.
- These are isomorphic: all have CRITICAL/HIGH/MEDIUM/LOW/INFO ± domain-specific values.
- A single ``FindingSeverity`` enum eliminates cross-type mapping and reduces import surface.

Advantages:

- One severity enum instead of three.
- New finding types automatically get severity without a new import.
- ``ContradictionSeverity`` mirrors ``FindingSeverity`` for contradictions (which are not findings but share the same severity scale).

Disadvantages:

- ``GapSeverity.BLOCKING`` and ``GapSeverity.RECOMMENDED`` are gap-specific values not shared by Observation or Risk. ``FindingSeverity`` must include all values, widening the enum beyond what any single finding type uses.
- Risk severity values (CRITICAL/HIGH/MEDIUM/LOW/INFO) are a subset of FindingSeverity — no issue.
- Observation has no severity in the baseline (it uses ``ObservationType`` instead). Mapping to severity adds expressiveness but changes the model.

Backward compatibility implications:

- ``GapSeverity.BLOCKING`` → ``FindingSeverity.BLOCKING``. Same string value, different type.
- Deprecated aliases for one cycle.

Effect on downstream components: Low — any reference to severity values uses string comparisons, which remain identical.

Complexity impact: DECREASED. 3 enums → 1 enum.

Recommendation: ACCEPT. Unified severity reduces enum surface. Provide deprecated aliases for one cycle.

Cross-Phase Impact

Phase	Δ1 Finding	Δ2 Contradiction	Δ3 ConfidenceScore	Δ4 Session	Δ5 Assumption	Δ6 Constraint	Δ7 Stale Rule	Δ8 Sever
Phase G (Planning)	Reads single list	Type-filtered contradictions	Reads confidence	Uses session for plan correlation	Reads assumptions for planning risk	Reads constraints for plan bounds	Detects stale before planning	N/A
Phase H (Governance)	Policy rules iterate once	Type-based policy routing	Confidence threshold for auto- approval	Session audit trail	Reviews assumptions for policy compliance	Validates constraints against policy	Freshness gate for governance	N/A
Phase I (Execution)	Neutral	Neutral	Neutral	Session tracking	Neutral	Constraint- enforced scheduling	Prevents stale execution	N/A
Phase J (Workflow)	Neutral	Neutral	Neutral	Session chaining	Neutral	Neutral	Neutral	N/A

Key insight: All accepted deltas are either neutral or positive for all downstream phases. No delta introduces a dependency that constrains Phase G, H, I, or J design.

Risk Assessment

Risk	Likelihood	Impact	Mitigation
External consumers break on key rename	Medium (depends on consumer count)	High	Deprecated aliases + one-cycle migration window
Persistence schema migration fails	Low (no persistent storage in Phase F)	Medium	Schema versioning + migration script with phase boundary
Event bus consumers miss events	Low (Phase F events are optional)	Low	Dual-publish during migration cycle
Computed properties have perf impact	Very Low (<200 items)	Negligible	Micro-benchmark before migration
Stale context rule false positive on batch jobs	Medium	Medium	Configurable threshold, not hardcoded
Assumption/Constraint lists always empty	High (until rules emit them)	Low	Documented as "ready for G/H" — zero defect

Overall risk: LOW. All changes are add-only or rename-only except $\Delta 7$ (stale rule, which is additive). No data migrations, no behavioural changes, no I/O pattern changes.

Compatibility Assessment

Dimension	Baseline	Proposed	Compatible?
Python type imports	Observation, Conflict, etc.	Deprecated aliases	✓ One-cycle transition
ReasoningResult.to_dict() keys	observations, conflicts, gaps, risks	findings, contradictions	✗ Breaking — needs schema versioning
RuleResult fields	4 separate lists	findings + contradictions	✗ Breaking — dual-support needed
ConfidenceEngine.assess() signature	4 params	2 params	✗ Breaking — dual-support needed
Metrics keys	conflicts_total	contradictions_total	✗ Need alias or migration
Event payload keys	observations, conflicts	findings, contradictions	✗ Schema versioning needed

Migration strategy: Dual-support pattern for one phase cycle — both old and new fields present on RuleResult and ReasoningResult. Deprecated aliases for class names. Deprecation warnings for old access patterns at runtime.

Long-Term Architectural Consequences

Decade-scale stability

The proposed architecture separates the reasoning model into two fundamental concepts:

1. **Findings** — any proposition about the context (observation, gap, risk, or future types).
2. **Contradictions** — any detected inconsistency between propositions (including stale context, incomplete evidence, duplicate findings).

This dichotomy (proposition + inconsistency) is general enough to accommodate any future reasoning requirement without structural refactoring. Adding "insight" or "recommendation" findings, or "trust violation" contradictions, requires only new enum values.

Downstream engine coupling

The canonical reasoning output (`findings` + `contradictions` + `assumptions` + `constraints`) forms a contract that downstream engines can depend on. Because it is additive (lists grow, fields never shrink), consumers can safely ignore types they do not handle. A Governance Engine in Phase H can ignore `assumptions` if it does not need them; a Planning Engine in Phase G can ignore `constraints` but benefit when it arrives.

Persistence schema

If `ReasoningResult` is stored to a time-series database (e.g., for auditing), the proposed schema (`findings[].finding_type`, `contradictions[].contradiction_type`) is easier to query than separate tables for observation/conflict/gap/risk. A single `WHERE finding_type = 'risk'` replaces a table join.

API surface stability

The proposed API surface is:

```
result = engine.evaluate(context)
result.findings      # List[Finding] – all propositions
result.observations  # computed – observations only
result.gaps          # computed – gaps only
result.risks         # computed – risks only
result.contradictions # List[Contradiction]
result.assumptions   # List[Assumption]
result.constraints    # List[Constraint]
result.confidence     # ConfidenceScore
```

This surface is stable across any future enumeration of finding types. Adding `result.insights` requires only a new computed property on `ReasoningResult` — no schema change.

Final Recommendation Matrix

Delta	Change	Recommendation	Justification
Δ1	Observation/Gap/Risk → Finding	ACCEPT	40% code surface reduction, computed properties preserve API, no behavioural change
Δ2	Conflict → Contradiction	ACCEPT	G5.5 alignment, typed discrimination enables downstream filtering
Δ3	ConfidenceAssessment → ConfidenceScore	ACCEPT	Terminological precision, deprecated alias provided
Δ4	ReasoningSession	DEFER	Zero production value in Phase F; implement when first multi-evaluation consumer arrives
Δ5	Assumption	ACCEPT	G5.5 requirement, add-only, enables governance audit of reasoning assumptions
Δ6	Constraint	ACCEPT	G5.5 requirement, add-only, downstream value in Phase G/H/I
Δ7	Extended Contradiction Detection	ACCEPT	Compliance gap fix — G5.5 §5 requires stale context detection; make threshold configurable
Δ8	Unified Severity	ACCEPT	3 enums → 1 enum, deprecated aliases provided

7 ACCEPT | 1 DEFER | 0 REJECT | 0 REQUIRES MODIFICATION

The sole deferral (Δ4 — ReasoningSession) is recommended for Phase G or H, when the first multi-evaluation workflow is designed.

Recommended Canonical Architecture

```
app/shunya/reasoning/models.py
```

```
Finding          # Unified: Observation | Gap | Risk
    finding_type: str # "observation" | "gap" | "risk"
    severity: str    # FindingSeverity value
    fact_key, fact_value, label, description, source
    confidence, evidence, metadata, created_at

Contradiction     # Renamed from Conflict
    contradiction_type: str # ContradictionType value
    severity: str         # ContradictionSeverity value
    fact_keys, finding_ids, sources, resolution_guidance
    evidence, metadata, created_at

Assumption        # NEW — explicit assumption tracking
Constraint        # NEW — explicit constraint tracking
ConfidenceScore   # Renamed from ConfidenceAssessment
EvidenceReference # Unchanged
ReasoningMetadata # Unchanged
ReasoningResult   # Uses findings, contradictions, assumptions, constraints
ReasoningSession  # DEFERRED — add in Phase G or H

FindingSeverity   # Unified: CRITICAL..BLOCKING..OPTIONAL
ContradictionSeverity # CRITICAL..INFO (same as old ConflictSeverity)
ContradictionType # fact_conflict, assumption_conflict, stale_context,
```

```
ConfidenceLevel      # incomplete_evidence, duplicate_finding
                     # Unchanged
```

With deprecated aliases for one phase cycle:

```
Observation = Finding      # with finding_type="observation"
Gap = Finding              # with finding_type="gap"
Risk = Finding             # with finding_type="risk"
Conflict = Contradiction
ConflictSeverity = ContradictionSeverity
ConfidenceAssessment = ConfidenceScore
```

Outstanding Questions

- 1. Stale context threshold:** Should the staleness threshold be configurable per-environment (e.g., 5 min for real-time, 24 h for batch)? Currently 1 hour is hardcoded. **Proposal:** Accept as-is, add configuration in Phase G.
- 2. Duplicate finding detection:** G5.5 §5 requires `duplicate_finding` detection. The refactored code added the `ContradictionType` enum value but no rule actually detects duplicates. Should this rule be added in Phase F or deferred to Phase G? **Proposal:** Defer to Phase G — duplicate detection requires inter-finding comparison across rule results, which is a cross-cutting concern better addressed after the Finding model is stable.
- 3. Backward compatibility window:** How many phase cycles should deprecated aliases be maintained? **Proposal:** One cycle (Phase G). Remove deprecated aliases at the start of Phase H.
- 4. Event schema versioning:** If events are published during the migration window, should the event bus dual-publish (`observations` AND `findings`)? **Proposal:** Dual-publish during migration cycle; remove old keys at Phase H.

Sign-Off Block

```
Architectural Impact Analysis Complete.

7 ACCEPT | 1 DEFER | 0 REJECT | 0 REQUIRES MODIFICATION

Awaiting Governance review.
```

End of Architectural Impact Analysis

